

Class 16: Social encounters and turning points

Matthew J. Salganik

Sociology 204: Social Networks
Princeton University

Wednesday, March 26, 2025



Laptop policy:

- ▶ Default: No laptops in lecture
- ▶ If you would like to use a laptop in lecture, please email me explaining your reasons so that we can learn from them. All reasonable requests will be accepted.
- ▶ All students using laptops must sit in the last row so as not to distract their classmates (again email me if you need a change).

Review:

- ▶ economic connectedness (but not other measures of social capital) are related to upward income mobility

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- ▶ this relationship appears to be causal
- ▶ economic connectedness is related to exposure and friending bias
- ▶ exposure and friending bias vary across settings and may be amenable to policy interventions

1. Bandura (1982). The Psychology of Chance Encounters and Life Paths. *American Psychologist*.
2. Goffman (2019). Go to More Parties? Social Occasions as Home to Unexpected Turning Points in Life Trajectories. *Social Psychology Quarterly*.



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A PHILOSOPHICAL ESSAY
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BY
PIERRE SIMON, MARQUIS DE LAPLACE.

TRANSLATED FROM THE SIXTH FRENCH EDITION

BY
FREDERICK WILSON TRUSCOTT, PH.D. (HARV.),
Professor of Germanic Languages in the West Virginia University,

AND
FREDERICK LINCOLN EMORY, M.E. (WOR. POLY. INST.),
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FIRST EDITION

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1902.

“An intellect which at a certain moment would know all forces that set nature in motion, and all positions of all items of which nature is composed, if this intellect were also vast enough to submit these data to analysis, it would embrace in a single formula the movements of the greatest bodies of the universe and those of the tiniest atom; for such an intellect nothing would be uncertain and the future just like the past could be present before its eyes.” Laplace

Opinion: Even if this is true, it is the wrong mental model. We would be better off if this idea died.

The Psychology of Chance Encounters and Life Paths

ALBERT BANDURA *Stanford University*

“The central thesis of this article is that chance encounters play a prominent role in shaping the course of human lives.”

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1. Do you think this is true?

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1. Do you think this is true?
2. Did this article change your mind?

- ▶ “The central thesis of this article is that chance encounters play a prominent role in shaping the course of human lives.”

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- ▶ “A comprehensive developmental theory must therefore specify factors that set and alter participate life courses if it is to be provided an adequate explanation of human behavior.”
- ▶ “. . . . a chance encounter is defined as an unintended meeting of persons unfamiliar to each other.”

Claims:

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1. "Psychology cannot foretell the occurrence of fortuitous encounters, however sophisticated its knowledge of human behavior becomes."
2. "However, psychological knowledge can provide the basis for predicting the nature, scope, and strength of the impact they will have on human lives."

- ▶ Personal determinants of the impact of chance encounters

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 - ▶ entry skills
 - ▶ emotional ties
 - ▶ values and personal standards

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Only evidences comes from extreme anecdotes

Go to More Parties? Social Occasions as Home to Unexpected Turning Points in Life Trajectories

Alice Goffman¹

Social Psychology Quarterly

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journals.sagepub.com/home/spq



In a 1981 presidential address to the Western Psychological Association, the psychologist Bandura (1982:749) noted that fortuitous encounters play a central role in shaping the course of human lives but that “psychology cannot foretell the occurrence of fortuitous encounters, however sophisticated its knowledge of human behavior becomes.” This sociological paper responds by identifying one concrete situation in which fortuitous encounters tend to emerge: social occasions.¹¹

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The answer proposed here is to closely study the moments in which bursts of change emerge.

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“The limitations of these 3 data sources are significant and not fully mitigated by combining them.”

course. That is, social occasions are more likely than other kinds of time to house events that unexpectedly shift the trajectory of individual lives. They do this by thrusting people into a special world, building collective effervescence and emotional energy, gathering usually dispersed people together, requiring that participants publicly rank their relations, and demanding complex choreography, carried out while others watch and judge. The more social occasions do these five things, the more likely they are to become unexpectedly consequential, opening people to changes in their bonds, habits, thinking, and plans.

What events do you have like this coming up?

- ▶ a special world set off from ordinary life
- ▶ collective effervescence and emotional energy
- ▶ worlds colliding
- ▶ forced public rankings
- ▶ complex choreography (while everyone is watching)

1980; Tavory 2009:61). Still, social occasions represent a pocket of fatefulness in everyday life, a form of experience more hospitable to bringing about some unanticipated change than many other things people do throughout the day.

Differently put, our trajectories through life vary in ways that cannot be fully explained by the characteristics of people because each one of us is subject to fateful situations: forms of experience that pull us out of our routines and open us to bursts of change going in many directions. The social occasion is one such form of experience and as such, merits attention by anyone interested in better explaining our stunningly varied life paths.

Life outcome prediction task: Given some data about a person and their environment, how well can we predict their future outcomes?

If LLMs are “auto-complete on steroids”, could we build something like this for life trajectories?

Measuring the predictability of life outcomes with a scientific mass collaboration

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FF **Fragile
Families**
& Child Wellbeing Study
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Future of Families & Child Wellbeing Study

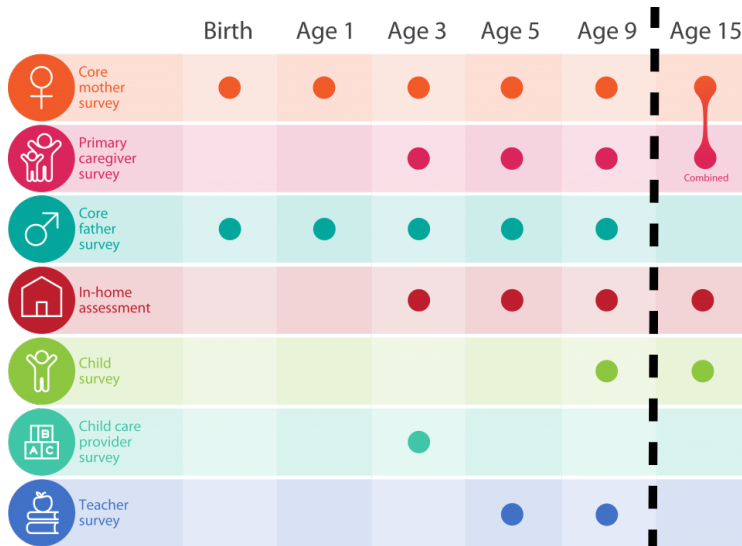
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Future of Families & Child Wellbeing Study

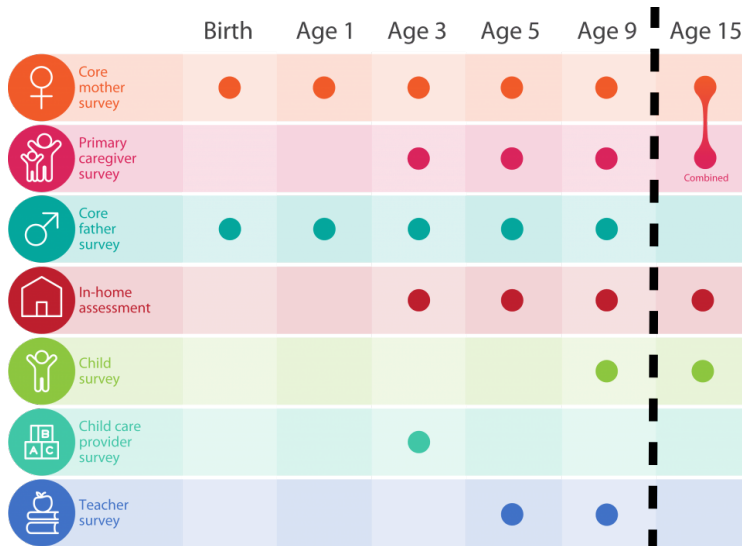
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- ▶ Thousands of children born in 20 U.S. cities with an over-sample of non-marital births
- ▶ Followed from birth
- ▶ Already used in more than 1,000 published papers



Data module	Child age	Module
Mother	Birth	Child health and development; Mother-father relationships; Fatherhood; Marriage attitudes; Relationship with extended kin; Environmental factors and government programs; Health and health behavior; Demographic characteristics; Education and employment; Income

Data module	Child age	Sections
Mother	Birth	A) Child health and development, B) Father-mother relationships, C) Fatherhood, D) Marriage attitudes, E) Relationship with extended kin, F) Environmental factors and government programs, G) Health and health behavior, H) Demographic characteristics, I) Education and employment, J) Income
Father	Birth	A) Child health and development, B) Father-mother relationships, C) Fatherhood, D) Marriage attitudes, E) Relationship with extended kin, F) Environmental factors and government programs, G) Health and health behavior, H) Demographic characteristics, I) Education and employment, J) Work activities, K) Income
Mother	1	A) Family characteristics, B) Child well-being and mothering, C) Father-child relationship, D) Mother's relationship with father, E) Current partner, F) Demographics, G) Mother's family background and support, H) Environment and programs, J) Health and health behavior, K) Education and employment, L) Income
Father	1	A) Family characteristics, B) Child well-being and fathering, C) Mother-child relationship, D) Father's relationship with mother, E) Current partner, F) Demographics, G) Father's family background and support, H) Environment and programs, J) Health and health behavior, K) Education and employment, L) Income
Mother	3	A) Family characteristics, B) Child well-being and mothering, C) Father-child relationship, D) Mother's relationship with father, E) Current partner, F) Demographics, H) Mother's family background and support, I) Environment and programs, J) Health and health behavior, K) Religion, L) Education and employment, L) Income
Father	3	A) Family characteristics, B) Child well-being and fathering, C) Mother-child relationship, D) Father's relationship with mother, E) Current partner, F) Demographics, H) Father's family background and support, I) Environment and programs, J) Health and health behavior, K) Religion, L) Education and employment, L) Income
Primary care giver and in-home observation	3	A) Health and accidents, B) Family routines, C) Home toy and activity items, D) Nutrition, E) Food expenditures, F) Housing/building characteristics, G) Parental stress, H) Parental mastery, I) Discipline, K) Inferior social control and social cohesion and trust, L) Exposure to violence, M) Child's behavior problems, P) Observation checklist, Q) Common areas, R) Interior of house or apartment, S) Child's appearance, T) Home scale, U) Child emotion and cooperation, V) Ending
In-home activities with child and mother	3	A) Height and weight, B) Child's Peabody Picture Vocabulary Test or TVIP, C) Walk-A-Line, D) Q-Sort, E) Mothers Peabody Picture Vocabulary Test or TVIP, F) Child Care/Employment History Calendar
Child Care Provider Survey (for center-based care)	3	A) Care provided at the center, B) Care provided for focus child (information from director or teacher), C) Care provided for focus child (information from teacher), E) Teacher-parent relationship, F) Teacher beliefs, G) About the children's teacher
Child Care Center Observations	3	No clear section headings but contents include: Space and furnishings, personal care routines, language-reasoning, activities, interaction, program structure, parents and staff
Family Care Provider Survey (for family-based care)	3	A) Care provided, B) Child care routine and program, D) Provider-parent relationship, E) Child care provider beliefs, F) About the child care provider
Family Care Provider Observations	3	No clear section headings but contents include: Space and furnishings for care and learning, basic care, language and reasoning, learning activities, social development
Child Care Study Post-Observation Form	3	A) Observation checklist, B) Common areas, C) Interior of building, D) Home scale, E) Post-visit rating by interviewer
Mother	5	A) Family characteristics, B) Child well-being and mothering, C) Father-child relationship, D) Mother's relationship with father (for mothers who are or were in a relationship) E) Current partner, F) Demographics, H) Mother's family background and support, I) Environment and programs J) Health and health behavior, K) Religion K) Education and employment, L) Income
Father	5	A) Family characteristics, B) Child well-being and fathering C) Mother-child relationship D) Father's relationship with mother (for fathers who are or were in a relationship), E) Current partner, F) Demographics, H) Father's family background and support, I) Environment and programs J) Health and health behavior, K) Religion K) Education and employment, L) Income
Primary care giver and in-home observation	5	A) Health and accidents, B) Family routines, C) Home toy and activity items, D) Nutrition, E) Housing/building characteristics, F) Parental stress and mastery, G) Discipline, H) Exposure to violence, J) CPS contact, K) Food expenditures, L) Child's behavior, N) Activities, P) Observation checklist, Q) Common areas, R) Interior of house or apartment, S) Child's appearance, T) Home scale, U) Child emotion and cooperation, V) Ending
In-home activities with child and mother	5	A) Weight/height, B) Peabody Picture Vocabulary Test with child, C) Woodcock-Johnson Letter-Word activity with child, D) Attention sustained task, E) Child care employment history calendar, F) Five-minute sample, G) Peabody Picture Vocabulary Test with mother
Teacher	5	A) Information specific to the participating child, B) Academic skills specific to the participating child, C) Classroom behavior and social skills specific to the participating child, D) Classroom characteristics, E) Class resources and activities, F) School climate and environment, G) General information about teacher
Mother	9	A) Core mother interview: Family characteristics, household roster, marital, and fertility history, B) Bio father contributions and resources, C) Mother's relationship with father, D) Current partner, E) Mother's family background and support, F) Environment and programs, G) Health and health behavior, H) Religion, I) Education and employment, J) Income, K) Secondary caregiver
Father	9	A) Core father interview: Family characteristics, household roster, marital, and fertility history, B) Bio mother and bio father contributions and resources, C) Father's relationship with mother, D) Current partner, E) Father's family background and support, F) Environment and programs, G) Health and health behavior, H) Religion, I) Education and employment, J) Income, K) Secondary caregiver
Primary care giver	9	A) Introduction to non-parental caregiver survey, B) Mother-child relationship, C) Father-child relationship, D) Demographics, E) Income, education, and employment, F) Health and well-being, G) Environment, H) Health and accidents, I) Family routines and home life, J) Nutrition, K) Parental stress and mastery, L) Child's education, M) Child's neighborhood
Interviewer observation	9	A) Observation checklist, B) Common areas, C) Interior of house or apartment, D) Child's appearance, E) Home scale, F) Child emotion and cooperation, G) Ending
Child	9	A) Parental supervision and relationship, B) Parental discipline, C) Sibling relationships, D) Routines, E) School, F) Early delinquency, G) Task completion and behavior, H) Health and safety, I) Closing
In-home activities with child and primary care-giver	9	No clear section headings but activities include: Consent, Child assessment (PPVT, Digit span, Woodcock-Johnson Tests 9 and 10), Primary care-giver self-administered questionnaire, Health measures, Saliva sample, Biological mother weight, Child weight/height, Primary caregiver open-ended responses
Teacher	9	A) General information, B) Classroom behavior and social skills specific to the participating child, C) Information specific to the participating child, D) Parent/guardian involvement, E) Classroom characteristics, F) School climate and environment, G) General information about teacher



4,242 families

Birth to age 9
12,942 variables

Information about child and family

Background data

Age 15
6 variables

Training

Leaderboard

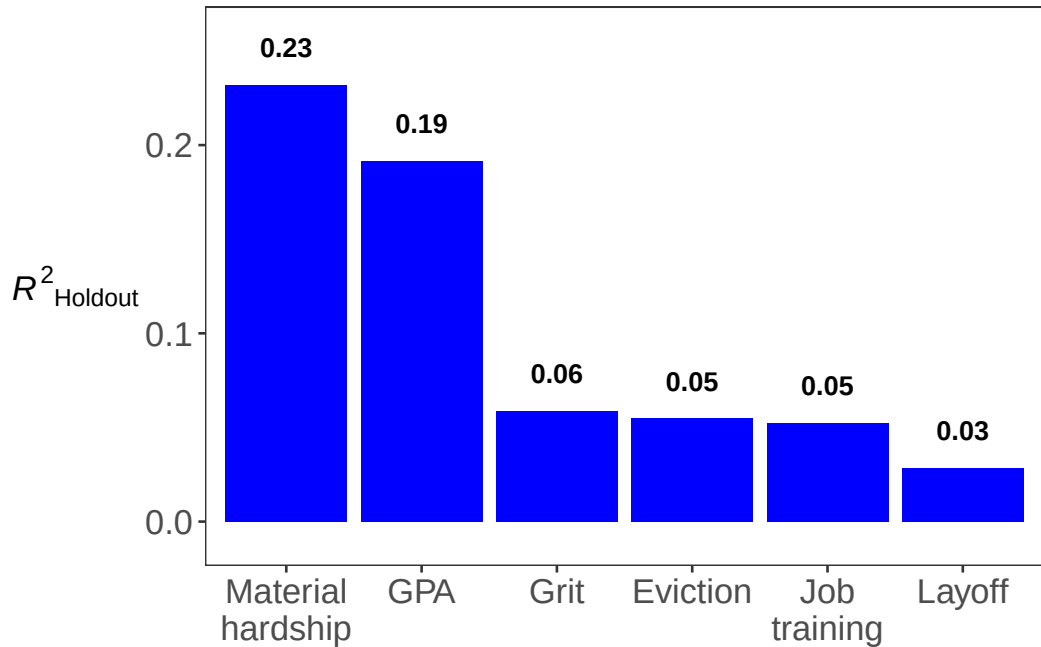
Holdout

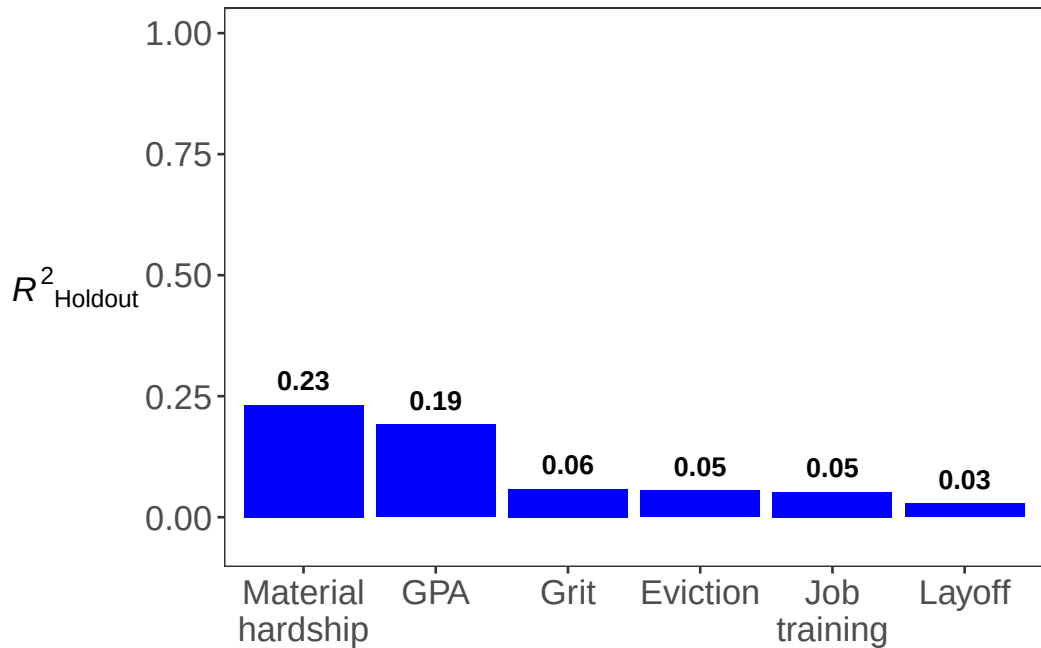
Outcome
data

Using a large, high-quality social science dataset collected since birth and modern machine learning methods, how accurately can we predict outcomes from children, parents, and families?

$$R^2_{holdout} = 1 - \frac{\sum_{i \in \text{holdout}} (\hat{y}_i - y_i)^2}{\sum_{i \in \text{holdout}} (\bar{y}_{train} - y_i)^2}$$

Six outcomes: Child grade point average (GPA), Child grit, Household eviction, Household material hardship, Adult job loss, Adult job training





What's going on?

CSML Machine Learning Lunchtime Seminar Series

Apr 22, 2025, 12:00 pm – 1:30 pm 📅

Bendheim House 103



Speaker

Matthew J. Salganik 
Princeton University

Details

Lunch is available beginning at 12 PM

Speaker to begin promptly at 12:30 PM

Abstract: TBD

Contributions to and/or sponsorship of any event does not constitute departmental or institutional endorsement of the specific program, speakers or views presented.



`https://csml.princeton.edu/events/
csml-machine-learning-lunchtime-seminar-series-4`

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Summary:

- ▶ sometimes life trajectories can evolve in unexpected ways
- ▶ we can try to understand when chance encounters will shift a life path (Bandura)
- ▶ and we can try to understand which kind of social occasions are more likely (Goffman)
- ▶ there is a lot about this that we still don't understand, but I hope that we can by using population-scale datasets and large language models (Salganik)

Have fun at precept this week.